AI BEST PRACTICES · ISM 6644 FINAL PROJECT

Stay in the driver's seat.
Don't let AI take the wheel.

I built our group's diabetes notebook with Claude Code, and it took about 27 inputs from me across three sessions.[E4] Section 6 is my main case: three typed prompts, three menu answers and one commit command.[E1]

My advice: the numbered headlines and the boxes tagged My advice.

Every other line comes from the project record and carries a key like this E1 into the evidence appendix.

CLAUDE DID THE TASKS[E6]

- Claude chose the Section 6 methods: 5 nearest neighbours, 1,000 label shuffles, seed 6644 and standardized inputs.[E6]
- In session 3 it turned my prompts into ten agent briefs.[E5] Across all three sessions, nine agent tasks worked on the notebook.[E5]
- Claude caught the hypertension leakage: the column matched target > 200 in all 442 rows.[E17]

I KEPT THE JUDGMENT[E6]

- I supplied the goal, the professor's requirements, the 15-minute limit and my pick from each menu.[E6]
- I decided how to fix the leakage.[E17] When the professor made the notebook the only deliverable, I changed the plan.[E18]
- In one menu I turned down the recommended option on two questions.[E19] Who presents what stayed with our group.[E20]

1 Give agents the tasks. Keep the judgment.

I never had to pick a nearest-neighbour setting, and Claude left who presents what to our group.[E6, E20]

2 Plan the work and the checks before you prompt.

My three Section 6 prompts came to 88 words, and none of them spelled out a method.[E2] The ten agent briefs Claude wrote in that session came to 3,060 words.[E2] One 44-word prompt plus one menu answer became the 549-word brief that set the similarity-graph methods.[E3]

That brief put the checks in before any code ran: print every number from code, plant a failure the checks must catch, and don't claim the profiles cluster unless the result beats the shuffled 95th percentile.[E7]

3 Treat an agent's report as a claim until it's rerun.

The main session itself rechecked two agents' results; after the hypertension fix, its rerun gave 36 cells and 0 errors.[E21] Checks were tested with planted failures they had to catch.[E22] When something wasn't rerun, the session said so plainly.[E23]

The main session questioned an agent's too-perfect split, 0% high-risk among patients without hypertension, before passing it on to me.[E25, E17]

4 A matching number can still mean the wrong thing.

A verifier agent found all 25 numbers it checked in the printed output.[E21, E24] My speaker notes still called 0.321 the chance level.[E24] It is the 95th percentile of the shuffled results; chance, the shuffled mean, is 0.301.[E11]

5 Try more ideas. Don't send 20 pages when 2 sentences do.

At 12:16 PM on Oct 4 I asked about 3D force-aware graphs in one 44-word prompt; by 12:22 PM Claude had written the brief for a 3D similarity graph built from 5 nearest neighbours.[E2, E3]

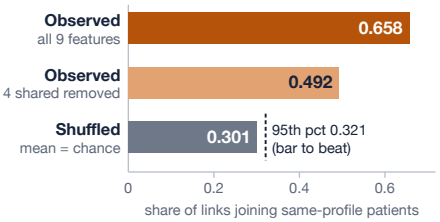
Two of our charts are as busy as the data: the similarity graph joins 442 patients with 1,578 links,[E8] and the parallel-coordinates chart draws one line per patient, so 442 lines.[E9] The printed share says it in one line: 0.658 of links join same-profile patients.[E10]

6 Let the harness carry the standards.

My rules' check instructions (No-claim, planted negative, verify) appear in four of the ten session-3 agent briefs.[E16] Advisory hooks that only reminded the session fired 42 times and drew no reaction.[E15]

At 12:39 PM on Oct 4 a guard blocked a command, and the main session reran it without the redirect instead of stopping to report.[E13] The rules treat a denial as a signal to stop, so that rerun may count as working around the guard.[E13] At 4:32 PM a second denial went the right way: the check went to a verifier agent.[E14]

Same-profile patients link up far more than chance[E10, E11]



In the similarity graph, 0.658 of links join patients in the same risk profile.[E10] Across 1,000 label shuffles the mean is 0.301, which is chance; 0.321 is the shuffled 95th percentile, the bar to beat.[E11] Four of the nine features also define the profiles; without them the share is still 0.492, against the same 0.301 mean.[E12]

MY ADVICE

Before you hit send

- ☐ Did I write down the goal, constraints and checks first?
- ☐ Can every number be traced to code or a source?
- ☐ Did I read the output for meaning, not just accept it?
- ☐ Does each chart or paragraph make one clear point?
- ☐ Is this the shortest version my audience needs?
- ☐ Which decisions here were mine, and do I stand behind them?

MY ADVICE

The driver's test

If you find yourself doing more of the work you enjoy and less of the work you dread, you're probably using AI correctly.

E-numbers point to code/final-project/ai-best-practices-evidence.md (source, locator and exact quote for each)